Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
D		2605710

Steering the Fever:

A Smarter, Safer Path to Wins and Value via Rolling MPC

Summary

A pro basketball franchise is a control problem, not a one-time decision: every season is a sequence of roster, finance, and marketing moves made under salary-cap rules and uncertainty, where "winning" and "making money" push and pull each other. For Indiana Fever in the WNBA, we build a **discrete-time rolling MPC framework** that links three state indices—competitive level, fan engagement, and financial status—to five controllable actions: leverage, cap utilization, star allocation, marketing, and ticket price. The model forecasts multi-window outcomes win rate, net profit, and valuation, then re-optimizes each decision window to keep the plan feasible and responsive.

For task I on roster adjustment, we use rolling MPC to output team-level targets at each decision window, then use a **lower-layer 0-1 roster optimizer** to translate targets into an implementable acquisition plan. The roster layer maximizes a **value-cost tradeoff** under salary cap, star ratio, roster size, and positional balance constraints. Comparing three acquisition styles, **Balanced** and **Aggressive** deliver stronger short-window improvements than **Conservative**; Conservative can spend more yet assemble weaker star composition and worsen the financial state, while Aggressive achieves the largest financial improvement with lower payroll, consistent with the policy comparison table.

For Task II on league expansion, we model expansion as a **time-varying disturbance** to the state dynamics plus a parameter shift that reshapes revenue–cost mapping. Expansion locations are evaluated via five interpretable indices (competition impact, derby premium, league growth, travel impact, and talent tightness), which jointly determine scenario differences; results show a consistent ranking where nearby domestic expansion generates the most negative net impact and far domestic expansion the most positive, and the controller correspondingly reallocates payroll/star/marketing effort to buffer shocks or exploit growth.

For Task III on player injury shocks, the injury is treated as a **one-time exogenous shock** to state variables, with performance loss computed by ΔWin and business loss decomposed into gate/media/merch channels (including attendance-driven gate revenue). MPC then re-optimizes with a risk-aware mean–variance design over short vs. long absence; the policy responds by tightening financial exposure and smoothing adjustments.

For Task IV on additional business decision—ticket pricing, we add ticket pricing as a new business control in the range 0.7 to 1.3 and link it to attendance and ticket revenue using a constant price elasticity. The season-long strategy is simple and executable: keep baseline pricing in 2026, then raise the average ticket price by **5 percent** per season from 2027 to 2029, and pause the next step if occupancy shows a clear drop, then re-optimize using the updated team state.

Overall, this rolling MPC model steers the Fever through a season of uncertainty by recalibrating the control variables at each decision window, delivering a smarter and safer path that balances wins and long-term value under hard league constraints.

Keywords: Model Predictive Control, Bilevel Optimization, Mean–Variance Optimization, Dynamic Pricing

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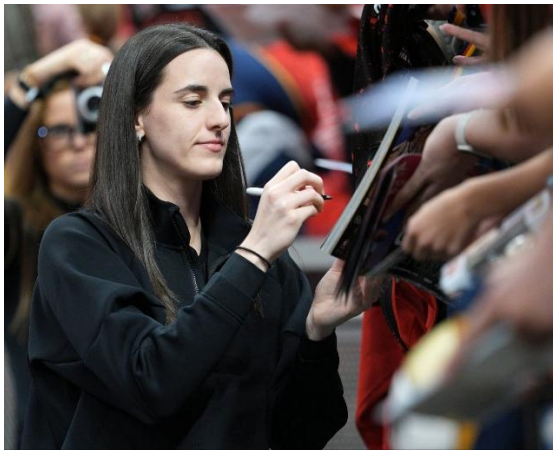
1 Introduction

1.1 Problem Background

Professional sports franchises operate as commercial systems in which on-court success is only one driver of value: while winning can amplify attention and media exposure, profitability also hinges on star power, timing, marketing, venue capacity, and local market conditions, all under constraints such as salary caps, roster limits, schedules, and shifting media-rights structures. In the WNBA, rising media visibility, expansion, potential arena relocations, and emerging digital platforms sharpen the trade-off between maximizing competitive performance and safeguarding financial health and long-term flexibility amid uncertainty. Because player value is inherently multidimensional and can change with injuries, economic conditions, and league reforms, effective team management requires an integrated decision framework that links sports analytics with financial modeling to support season planning, roster and contract decisions, player acquisition, and commercial operations.



(a) 2026 WNBA Season Schedule



(b) Popular Players' Interaction with Fans

Figure 1: WNBA Seasonal Exposure and Star-Driven Fan Engagement

1.2 Our work

Our work mainly includes:

✧ Establish a discrete-time MPC framework for the Indiana Fever that links team states with controllable management actions under practical constraints.

✧ Within the same MPC framework, we develop decision modules for baseline seasonal planning, bilevel roster acquisition , league expansion scenario adjustment, and injury-shock response, and further extend the model by adding ticket pricing as a new control to derive a season-long pricing strategy via constrained rolling optimization.

In summary, the whole modeling process can be shown as follows:

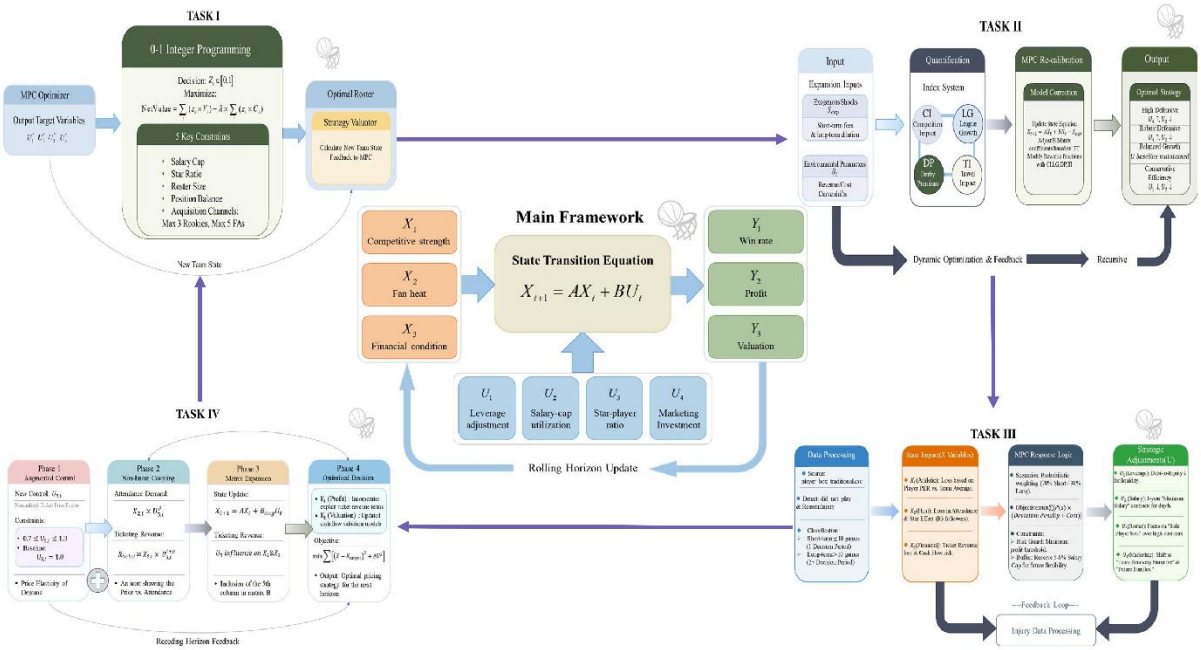


Figure 2: Our Work

2 Assumptions and Justifications

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

- **Assumption 1: Decisions are made at discrete "decision windows," and all variables are piecewise constant within each window.**

  Justification: Transaction windows and seasonal checkpoints naturally discretize management actions while facilitating the solution of equation results.

- **Assumption 2: The MPC uses $N = 4$ as the multi-season planning horizon to implement a rolling-horizon strategy with time-updating decisions.**

  Justification: A four-window horizon balances forward-looking planning and computational simplicity, and it is long enough to reflect multi-season effects while still allowing frequent re-optimization as new information arrives.

3 Notations

To avoid redundancy and maintain clarity, we do not provide a unified notation table here; instead, each section introduces its own symbols and variables with brief definitions when they first appear, since different tasks use different settings and parameterizations

4 Model Preparation

4.1 League and team selection

We anchor the study in the **WNBA** and use the **Indiana Fever** as the focal franchise.

4.2 Data sources

The data we use primarily comes from relevant public data provided by various websites. The data sources are summarized in Table 1.

Table 1: Notations used in this paper

Database Websites Data	Use
statista.com	Team financial data
espn.com	Team game information
trends.google.com	Team popularity
fever.wnba.com	Official team information
herhoopstats.com	Team salary data
stats.wnba.com	Official league game statistics
instagram.com	Team fan engagement data
wnba.com	Official league news and updates
pitchbook.com	Team financial data
kaggle.com	Team financial data

4.3 Practical note on data limitations

In practice, WNBA-related information is not delivered as a single clean dataset. Some sources update slowly, some revise historical entries without notice, and many pages expose only a narrow slice of what we would ideally measure.

5 Main Framework: Model Predictive Control

5.1 Why MPC

Team management in this task is a multi-objective, constraint-driven process that must be repeatedly updated across trade windows: decisions must balance competitive goals and business goals under salary-cap, contract, and leverage risks, while responding to performance deviations and external shocks. We therefore model operations as a discrete-time dynamic system in which the state evolves over time, management chooses controls each window, and these controls determine next-period outcomes. **Model Predictive Control (MPC)** fits this structure because it optimizes under constraints with a rolling horizon: it forecasts multi-period outcomes from the current

state, computes an action sequence, executes the first action, then updates and re-optimizes—forming a feedback loop that matches the task's rolling decision requirement and supports modular extensions.

5.2 Definition of State Variables, Control Variables, and Output Variables.

We index each trade window (or decision checkpoint) by discrete time t. The model uses:

- a **state vector** $X_t \in \mathbb{R}^3$,
- a **control vector** $U_t \in \mathbb{R}^4$,
- an **output vector** $Y_{t+1} \in \mathbb{R}^3$.

(1) State Variables X_t

$$X_t = \begin{bmatrix} X_{1,t} \\ X_{2,t} \\ X_{3,t} \end{bmatrix} \tag{1}$$

- $X_{1,t}$ **Team Competitive Strength:**
- $X_{2,t}$ **Fan Heat Index:**
- $X_{3,t}$ **Financial Condition Indicator:**

(2) Control Variables U_t

$$U_t = \begin{bmatrix} U_{1,t} \\ U_{2,t} \\ U_{3,t} \\ U_{4,t} \end{bmatrix} \tag{2}$$

- $U_{1,t}$ **Leverage Adjustment:** financing/repayment decisions; positive values increase leverage, negative values reduce leverage.
- $U_{2,t}$ **Salary Cap Utilization:** the proportion of total payroll to the league salary cap. Due to the league's strict cap enforcement and the substantial tax penalties incurred when exceeding the cap, we impose a hard constraint $0 \le U_{2,t} \le 1$
- $U_{3,t}$ **Star Player Ratio:** the proportion of star players within the lineup.
- $U_{4,t}$ **Marketing Investment:** expenditures allocated to brand development and promotional activities aimed at enhancing the team's visibility.

(3) Outputs Variables Y_t

$$Y_t = \begin{bmatrix} Y_{1,t} \\ Y_{2,t} \\ Y_{3,t} \end{bmatrix} \tag{3}$$

- $Y_{1,t}$ **Win Rate:** the primary competitive performance metric.

- $Y_{2,t}$ **Net Profit:** revenue minus major expenses (compensation and marketing).
- $Y_{3,t}$ **Franchise Valuation:** the long-term value indicator driven by profitability and brand equity.

5.3 Quantification Rules for Model Variables O_{Guard}

5.3.1 Quantifying Team Competitive Strength X_1

5.3.1.1 Quantifying Player Competitive Skil

We convert player-level information into reproducible numeric scores by building a position-aware, five-dimension evaluation system (offense, defense, availability/endurance, team contribution, and clutch performance), using position-specific indicators and weights to ensure interpretability and comparability across players.

(1) Dimension Definitions and Overall Framework

For any player i ,given her position $pos \in \{Guard, Forward, Center\}$, we compute five sub-scores:

$$O_i\ , D_i\ , Ph_i\ , T_i\ , C_i \in [0,1] \tag{4}$$

and then aggregate them into an overall competitive skill score P_i (used as the player's "strength" measure):

$$P_i = w_1 O_i + w_2 D_i + w_3 Ph_i + w_4 T_i + w_5 C_i \tag{5}$$

where the weights ω_k depend on position, reflecting position-specific preferences.

(2) Guard Evaluation System

We quantify guard performance using five standardized dimensions (offense, defense, availability, team contribution, and clutch impact) computed from public box-score/advanced metrics and aggregate them into a single guard index P_{Guard} via a weighted sum.

Table 2: Guard evaluation metrics and aggregation

Symbol	Metric definition
O_{Guard}	$O_{Guard} = 0.3 \cdot AST_TO + 0.25 \cdot AST_Share + 0.3 \cdot TS\% + 0.1 \cdot 3PT_Rate + 0.05 \cdot 3PT\%.$
D_{Guard}	$D_{Guard} = 0.60 \cdot Steal_Rate + 0.40 \cdot Adjusted_PM$
Ph_{Guard}	$. Ph_{Guard} = 0.70 \cdot Minutes_Stability + 0.30 \cdot max(0, B2B_Impact)$.
T_{Guard}	$T_{Guard} = 0.40 \cdot Network_Centrality + 0.60 \cdot Total_Impact$
C_{Guard}	$C_{Guard} = 0.70 \cdot Clutch_TS\% + 0.30 \cdot Clutch_Volume.$

Each component targets a distinct source of guard value.We combine the five dimension

scores into an overall guard index using a weighted sum:

$$P_{Guard} = 0.35O + 0.20D + 0.15Ph + 0.20T + 0.10C. \tag{6}$$

(3) Forward Evaluation System

We evaluate forwards using the same five-dimension scoring pipeline as guards by normalizing public box-score and advanced metrics to a common scale and aggregating them into a single forward index $P_{Forward}$ via a weighted sum.

Table 3: Forward evaluation metrics and aggregation

Symbol	Metric definition
$O_{Forward}$	$O_{Forward} = 0.40 \cdot Paint_Share + 0.30 \cdot OREB_Rate + 0.30 \cdot TS\%.$
$D_{Forward}$	$D_{Forward} = 0.60 \cdot Help_Defense_Rate + 0.40 \cdot Block_Rate$
$Ph_{Forward}$	$Ph_{Forward} = REB_Share$
T_{Forward}	same definition as in the guard system
$C_{Forward}$	same definition as in the guard system

The overall forward index is computed as

$$P_{Forward} = 0.30O + 0.25D + 0.20Ph + 0.15T + 0.10C \tag{7}$$

(4) Center Evaluation System

We evaluate centers using the same five-dimension structure as guards, while shifting the feature focus toward interior finishing, rebounding control, and rim protection. All indicators are standardized to a common scale and aligned so that higher values always indicate better performance. The five dimension scores are then aggregated into a single center index P_{Center} by a weighted sum.

Table 4: Center evaluation metrics and aggregation

Symbol	Metric definition
O_{Center}	$O_{Center} = 0.45 \cdot Paint_FG\% + 0.40 \cdot REB_Share + 0.15 \cdot High_Post_AST$
D_{Center}	$D_{Center} = 0.40 \cdot Block_Rate + 0.35 \cdot Deterrence + 0.25 \cdot Paint_Defense$
Ph_{Center}	$Ph_{Center} = \frac{rebounds}{fouls + rebounds}$
T_{Center}	same definition as in the guard system
C_{Center}	same definition as in the guard system

Each component targets a distinct source of center value.We combine the five dimension scores into an overall center index using a weighted sum:

$$P_{Center} = 0.25O + 0.35D + 0.20Ph + 0.15T + 0.05C \tag{8}$$

(5) Normalization and Direction Consistency

All indicators must ultimately undergo normalization.

5.3.1.2 Correlation Analysis

After defining player competitive strength, we examine whether it aligns with team outcomes by aggregating player scores to the team level using both a roster-wide average and a minutes-weighted average that emphasizes the core rotation. The results show a consistent positive relationship with win–loss outcomes and point differential, with the minutes-weighted measure exhibiting the stronger association, supporting competitive strength as a practical indicator of on-court performance for roster decision-making under salary-cap constraints.

Table 5: Center evaluation metrics and aggregation

Strength Metric	Outcome	r	p-value
Unweighted player strength	Win/Loss (point-biserial)	0.1919	0.0321
Minutes-weighted player strength	Win/Loss (point-biserial)	0.2349	0.0084
Minutes-weighted player strength	Point differential	0.2565	0.0039

5.3.1.3 Computation Rule for Team Competitive Strength

To adapt the team competitive strength X_1 for MPC state updates, we employ a weighted scoring system composed of four components.

(a) Playoff-weighted win rate component

Because playoff games are fewer but more informative, we upweight postseason outcomes by defining a playoff-weighted mixing coefficient based on regular-season and postseason game counts and an importance factor ($\kappa = 2$), then use this coefficient to form the combined win-rate score.

$$\lambda = \frac{\kappa GP}{GR + \kappa GP} \quad , \quad WR_{total} = (1-\lambda)WR_R + \lambda WR_P \tag{9}$$

(b) Efficiency and margin components

We use three additional components that capture scoring production, scoring prevention, and overall dominance:

- Offensive Rating (ORtg)
- Defensive Rating (DRtg)
- Net performance indicator (Net), implemented as net $net\ rating = \mathrm{ORtg} - \mathrm{DRtg}$

(c) Normalization

Since the four components are on different scales, each raw component is mapped to $[0,1]$ via Min–Max normalization using league-wide benchmarks for the same season.

(d) Composite score and weights

Let $WR_{norm}, ORtg_{norm}, DRtg_{norm}, Net_{norm}$ denote the normalized component scores. The final

team competitive strength is computed as:

$$X_1 = 0.40WR_{norm} + 0.25ORtg_{norm} + 0.25DRtg_{norm} + 0.10Net_{norm} \quad , \quad X_1 \in [0,1] \tag{10}$$

5.3.2 Quantifying Fan Heat Index X_2

5.3.2.1 Design principle and definition

As shown in Figures 2–3, team-level search interest is not evenly distributed across the roster; instead, it is highly concentrated around a few star players. Moreover, the "Indiana Fever" team keyword rises and falls largely in step with the spikes of those stars, indicating that superstar attention can directly translate into broader team visibility.

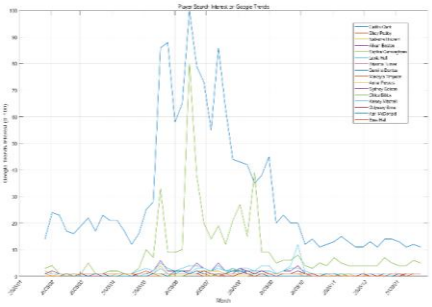


Figure 3: Player Search Interest on Google Trends

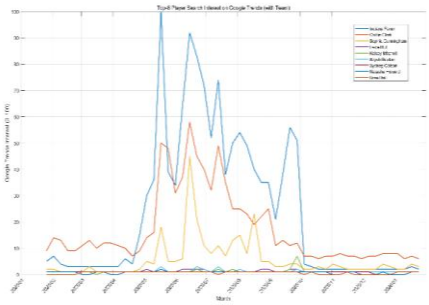


Figure 4: Top-8 Player Search Interest on Google Trends (with Team)

Fan Heat Index X_2 measures a team's market attention and fan engagement on a unified $[0,1]$ scale. We construct X_2 as a weighted composite of five dimensions that cover both offline demand and online exposure, while separating short-term momentum from long-run brand strength.

We define

$$X_2 = w_{att}S_{att} + w_{sm}S_{sm} + w_{mv}S_{mv} + w_{brand}S_{brand} + w_{star}S_{star} \tag{11}$$

with weights

$$(w_{att}, w_{sm}, w_{mv}, w_{brand}, w_{star}) = (0.35, 0.3, 0.2, 0.1, 0.05)$$

These weights prioritize direct participation and scalable attention, while keeping star-driven "extra heat" as a small adjustment rather than the main driver.

5.3.2.2 Dimension construction

We construct five component scores that capture fan demand, online attention, monetization strength, long-run brand position, and star-driven lift. Each component is computed from simple ratios or growth rates and then mapped to a common scale before aggregation into X_2. Table 6 summarizes the formulas.

Table 6: Fan heat index component definitions

Component	Formula
Attendance score S_{att}	$S_{att} = 0.7S_{rate} + 0.3S_{growth}$

Attendance rate S_{rate}	$S_{rate} = \frac{Average\ Attendance}{Arena\ Capacity}$
Attendance growth S_{growth}	$S_{growth} = \frac{A_t - A_{t-1}}{A_{t-1}}$
Social media score S_{sm}	$S_{sm} = 0.6S_{ratio} + 0.4S_{fg}$
Followers ratio S_{ratio}	$\mathrm{S}_{\mathrm{ratio}} = \frac{F_{team}}{F_{league_avg}}$
Followers growth S_{fg}	$\mathrm{S}_{fg} = \frac{F_t - F_{t-1}}{F_{t-1}}$
Market value score S_{mv}	$S_{mv} = 0.5S_{price} + 0.5S_{revenue}$
Ticket price ratio S_{price}	$S_{price} = \frac{P_{team}}{P_{league_avg}}$
Revenue share $S_{revenue}$	$S_{revenue} = \frac{R_{team}}{\sum_{teams} R}$
Brand awareness score S_{brand}	$S_{brand} = \frac{N - Rank + 1}{N}$
Star effect score S_{star}	$S_{star} = \frac{Recognition + Impact\ Approval}{200}$

Notation. In Table 6, A_t denotes the team's average home attendance in season t. F_t denotes the team's social media follower count in season t; F_{team} is the current follower level of the team and F_{league_avg} is the league average. P_{team} denotes the team's average ticket price and P_{league_avg} is the league average; R_{team} denotes team revenue and $\sum_{teams} R$ is total league revenue. N is the number of teams in the league, and $Rank$ is the team's valuation rank.

The final $X_2 \in [0,1]$ is used as the MPC state for fan heat.

5.3.3 Quantifying Financial Condition Indicator X_3

5.3.3.1 Design principle and definition

To represent a team's financial condition on a unified $[0,1]$ scale, we define X_3 as a weighted composite of four interpretable components: profitability, leverage risk, asset efficiency, and salary-structure discipline.

5.3.3.2 Component construction and formulas

Table 7: Financial condition indicator components

Component	Formula
Profitability F_1	$F_1 = \frac{R - C_s}{V}$
Total salary cost C_s	$C_s = \frac{\bar{S}N}{1000}$
Leverage risk F_2	$F_2 = -\frac{D}{V}$
Asset efficiency F_3	$F_3 = \frac{R}{V}$
Salary deviation F_4	$F_4 = -\frac{\lvert \bar{S} - \bar{S}_L \rvert}{\bar{S}_L}$
League average salary $\bar{S}$	$\bar{S}_L = \frac{1}{M}\sum_{j=1}^{M} \bar{S}_j$

Notation. R is team annual revenue, V is franchise valuation, D is total debt, $\bar{S}$ is team average player salary, N is roster size, and M is the number of teams. All indicators must be normalized.

5.3.3.3 Formula for Calculating X_3

We define

$$X_3 = \sum_{i=1}^{4} w_i F_{i,\text{norm}} \tag{12}$$

with weights

$$(w_1, w_2, w_3, w_4) = (0.35, 0.25, 0.25, 0.15)$$

We set the weights to reflect financial priorities in team operations: profitability is the primary sustainability signal, leverage and asset efficiency jointly capture risk and operational quality, and salary deviation is a secondary discipline term.

5.3.4 Control Variables

5.3.4.1 Design principle and definition

The control vector U_t represents the managerial levers that can be adjusted at each decision window. To keep the MPC implementable and aligned with league practice, each control is bounded within a feasible range that reflects regulatory limits or realistic operational capacity.

5.3.4.2 Feasibility bounds and parameter choices

We set an initial control vector U_0 using league benchmarks and team-level context, and impose feasibility bounds to ensure decisions remain realistic and compliant. In particular, the salary-cap utilization is strictly bounded by the league cap system, while the other controls are bounded by typical operating ranges observed in professional sports finance and WNBA market practice.

Table 8: Control-variable ranges and justifications

Control	Meaning	Initial value
U_1	Leverage level	0.18
U_2	Salary cap utilization	0.95
U_3	Star player ratio	0.71
U_4	Marketing investment, million USD	3.0

WNBA teams typically operate under conservative leverage, with debt-to-equity style ratios often in the 0.15–0.25 band.[1] At the same time, in subsequent practical operations, we will maintain U4 above 1 million to align with actual circumstances.

5.4 State Space Equation

To enable tractable rolling optimization with clear interpretation, we approximate team operational dynamics across decision windows using a linear state-space model:

$$X_{t+1} = AX_t + BU_t \tag{13}$$

where

$A \in \mathbb{R}^{3\times3}$ is the state transition matrix capturing inertia and cross-state coupling,

$B \in \mathbb{R}^{3\times4}$ is the control influence matrix capturing marginal effects of management actions,

5.5 Estimation of Matrix A

Matrix A reflects the natural evolution of the team's operational state under no intervention, along with the interactions between different states. We assign the value of each element as follows:

$$A = \begin{bmatrix} 0.45 & 0.08 & 0.04 \\ 0.35 & 0.75 & 0.10 \\ 0.2 & 0.15 & 1.02 \end{bmatrix}$$

5.6 Estimation of Matrix B

Matrix B encodes the marginal effects of management control on the next period's window state, where the value of each element is specifically defined as:

$$B = \begin{bmatrix} 0.03 & 0.42 & 0.08 & 0.01 \\ 0.01 & 0.12 & 0.95 & 0.28 \\ -20 & -0.18 & -0.08 & -0.05 \end{bmatrix}$$

5.7 Output Mapping

We map business outcomes from the predicted states and management decisions using simple, interpretable linear relationships. Let the state be $X_t = [X_{1,t}, X_{2,t}, X_{3,t}]^T$ and the control be $U_t = [U_{1,t}, U_{2,t}, U_{3,t}, U_{4,t}]^T$. The output vector is $Y_{t+1} = [Y_{1,t+1}, Y_{4,t+1}, Y_{5,t+1}]^T$

(1) Win Rate

$$Y_{1,t+1} = 0.5X_{1,t} + 0.2U_{2,t} + 0.15U_{3,t} \tag{14}$$

(2) Net Profit

$$Y_{2,t+1} = 3.5X_{2,t} - 0.66U_{2,t} + U_{4,t} - 0.05X_{3,t} \tag{15}$$

(3) Franchise Valuation

Valuation reflects long-run cash-flow potential and brand equity, proxied by profit and fan heat:

$$Y_{3,t+1} = 20X_{1,t} - 8X_{3,t} + 12U_{2,t} + 24U_{4,t} \tag{16}$$

This yields an interpretable causal chain:

$$U_t \to X_{t+1} \to Y_{t+1}$$

5.8 Multi-Objective Tracking with Smooth Policies

We adopt a receding-horizon MPC policy. At each window t, we choose a control sequence $\{U_t, \ldots, U_{t+N-1}\}$ to minimize deviation from management targets while discouraging abrupt policy shifts.

$$J_t = \sum_{k=1}^{N} \left[(Y_{t+k} - W)^T Q (Y_{t+k} - W) + (\Delta U_{t+k-1})^T R (\Delta U_{t+k-1}) \right] \tag{17}$$

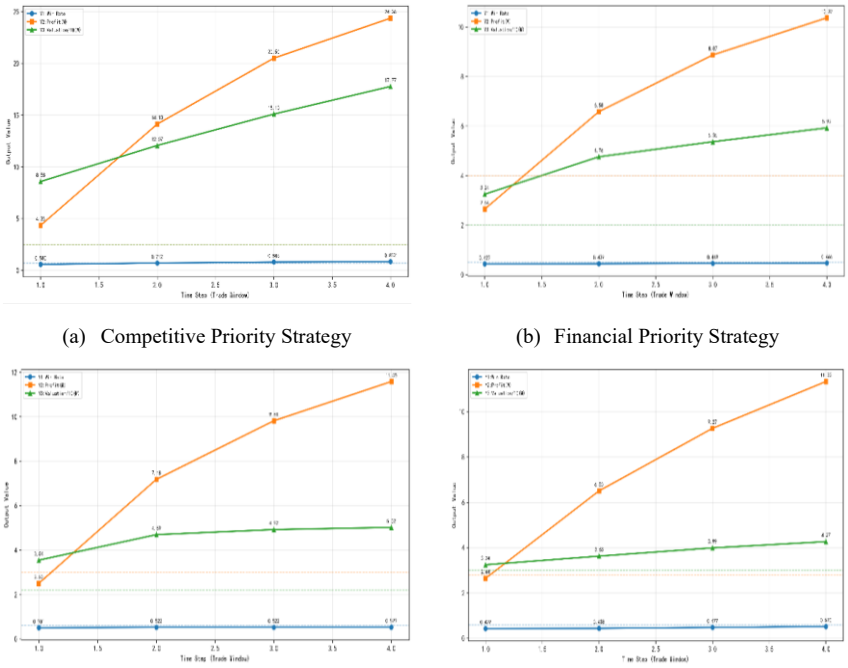
Constraints：

$$U_{\min} \le U_{t+k} \le U_{\max}, \qquad |\Delta U_{t+k}| \le \Delta U_{\max} \tag{18}$$

Here, $\Delta U_t = U_t - U_{t-1}$. $W = [W_1, W_4, W_5]^T$ is the target vector for win rate, net profit, and valuation; Q weights the importance of hitting each target; R penalizes large policy jumps to keep decisions implementable and stable. Only the first control $U_t^{\star}$ is executed, then the horizon rolls forward and the problem is re-solved at $t+1$.

5.9 Results and Analysis

In the simulation, we use a four-window horizon and treat the league environment as fixed relative to 2025. To reflect different ownership preferences, we evaluate four strategic styles—Competitive, Financial, Balanced, and Valuation growth—each represented by distinct settings of salary-cap utilization, star allocation, leverage, and marketing intensity; the resulting trajectories of the output vector components are summarized in Figure 5.



(a) Competitive Priority Strategy

(b) Financial Priority Strategy

(c) Balanced Development Strategy

(d) Valuation Growth Strategy

Figure 5: Predicted output trajectories for the four scenarios

Table 9-12 below present the specific values of control variables for the following four seasons to demonstrate strategy selection.

Table 9: Control Sequence Planning Under Competitive Priority Strategy

Year	U_1	U_2	U_3	U_4
2026	0.18	0.95	0.71	3
2027	0.18	0.95	0.71	3
2028	0.18	0.95	0.71	3
2029	0.18	0.95	0.71	3

Table 10: Control Sequence Planning Under Financial Priority Strategy

Year	U_1	U_2	U_3	U_4
2026	0.095	0.5	0.3	1
2027	0.045	0.5	0.3	1
2028	0.045	0.5	0.3	1
2029	0.02	0.5	0.3	1

Table 11: Control Sequence Planning Under Balanced Development Strategy

Year	U_1	U_2	U_3	U_4
2026	0.065	0.7317	0.4763	1

2027	0.0289	0.6243	0.3184	1
2028	0.0229	0.5041	0.3091	1
2029	0.0141	0.5020	0.3057	1

Table 12: Control Sequence Planning Under Valuation Growth Strategy

Year	U_1	U_2	U_3	U_4
2026	0.065	0.7317	0.4763	1
2027	0.0289	0.6243	0.3184	1
2028	0.0229	0.5041	0.3091	1
2029	0.0141	0.5020	0.3057	1

Using a rolling-horizon MPC, we generate implementable multi-window policies under four strategic preferences—**Competitive** (higher U_2, U_3 for wins), **Financial** (lower-cost controls for profit), **Balanced** (moderate trade-offs), and **Valuation growth** (actions that support long-run value)—providing a forward-looking link from priorities to actions to outcomes.

6 Task I: Player Selection for the Next Quarter

The baseline MPC produces optimal aggregate operating targets $U_t^* = [U_{1,t}^*, U_{2,t}^*, U_{3,t}^*, U_{6,t}^*]^T$ but it does not specify which players should be acquired to realize those targets. To bridge this "control-to-roster" gap with minimal structural change, we add a lower-layer roster acquisition module that translates MPC's targets into a concrete player set under league rules. The upper-layer MPC remains unchanged; only an actionable mapping from U^* to player decisions is added.

6.1 Two-layer decision architecture

We use a two-level structure: an upper-layer MPC computes team-level target controls U_t^* from the state-space dynamics, and a lower-layer 0–1 roster optimizer selects a feasible player set from the candidate pool to best realize U_t^*.

6.2 Candidate-pool construction and data augmentation

Because official channels do not provide complete information on new players, we construct a candidate pool and solve a 0–1 roster selection problem that chooses a feasible set of players matching the MPC targets $U_{2,t}^*, U_{3,t}^*$ while satisfying roster-size and position-balance constraints.

The pool set $\mathcal{I}$ consists of two standard acquisition channels:

- **Free agency set $\mathcal{I}^{FA}$:** higher-ready talent, higher cost.
- **Draft/rookie set $\mathcal{I}^{R}$:** low cost with higher uncertainty.

6.3 Lower-layer 0–1 roster acquisition mode

Given the MPC outputs for the next window, we translate policy targets into a discrete acquisition decision: selecting a subset of candidates that maximizes roster value while respecting payroll and composition constraints.

6.3.1 Decision variables

Let $z_i \in [0,1]$ indicate whether candidate $i \in \mathcal{I}$ is selected for acquisition/retention in the next window.

6.3.2 Objective: maximize net roster value

We maximize a value–cost tradeoff:

$$\max_z \sum_{i\in\mathcal{I}} z_i V_i - \lambda \sum_{i\in\mathcal{I}} z_i C_i \tag{19}$$

where V_i is the composite player value score and C_i is the salary. $\lambda > 0$ tunes cost sensitivity.

6.3.3 Core feasibility constraints

Table 13: Lower-layer feasibility constraints

Constraint type	Expression
Payroll cap	$\sum_{i\in\mathcal{I}} z_i C_i \le \mathrm{CAP}\cdot U^*_{2,t}$
Star ratio	$\frac{\sum_{i\in\mathcal{I}} z_i \mathrm{star}_i}{\sum_{i\in\mathcal{I}} z_i} \ge U^*_{3,t}$
Roster size	$12 \le \sum_{i\in\mathcal{I}} z_i \le 15$

6.4 Feedback: converting roster decisions back to team states

After solving the lower-layer problem, we compute the achieved control levels and next-state estimates, which closes the loop:

- Achieved payroll utilization:

$$\hat{U}_{2,t} = \frac{\sum_i z_i C_i}{\mathrm{CAP}_t} \tag{20}$$

- Achieved star ratio:

$$\hat{U}_{3,t} = \frac{\sum_i z_i, \mathrm{star}_i}{\sum_i z_i} \tag{21}$$

We then update the team state proxies:

$$\begin{aligned} X^{new}_{1,t} &= \mathrm{Agg}\left(x_{1,i}\cdot z_i = 1\right), \\ X^{new}_{2,t} &= \mathrm{Agg}\left(x_{2,i}\cdot z_i = 1\right) + \mathrm{StarBoost}, \\ X^{new}_{3,t} &= X_{3,t} - \kappa \sum_i z_i C_i + \mathrm{RevGain} \end{aligned} \tag{22}$$

6.5 Results and analysis

We compare three acquisition styles—**Conservative**, **Balanced**, and **Aggressive**—that differ in risk appetite and resource concentration, and evaluate them by roster/payroll feasibility and the implied one-window state update (X_1, X_2, X_3) relative to the 2025 baseline.

Table 14: Summary of roster outputs and state changes

Scenario	Players	Stars	Total payroll (M$)	ΔX_1 (%)	ΔX_2 (%)	ΔX_3 (%)
Conservative	13	6	2.080	+43.4	+42.24	−7.2
Balanced	12	7	1.557	+56.6	+73.1	+35.8
Aggressive	12	6	1.340	+57.4	+71.8	+53.7

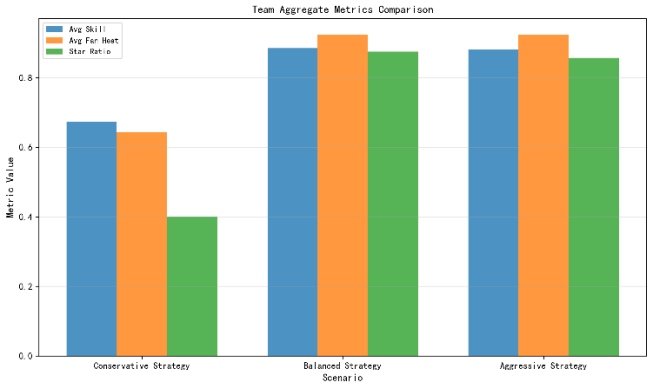


Figure 6: Team Aggregate Metrics Comparison

Overall, the Balanced and Aggressive policies deliver similarly strong short-horizon gains in on-court quality and fan attention, while they diverge on financial impact: the Aggressive plan achieves the largest improvement in financial state ith the lowest payroll, whereas the Balanced plan slightly leads in fan-heat growthat moderate payroll ; by contrast, the Conservative plan is the most payroll-intensive yet yields the weakest roster "star ratio" and the only deterioration in financial condition, indicating that conservative acquisition here trades away financial sustainability without matching gains in competitive/attention outcomes.

7 Task II: League Expansion

7.1 Overall frame

League expansion is incorporated into our MPC framework through two mechanisms: a time-varying exogenous shock $S_{\text{exp}}(t)$ that directly perturbs the team state vector, and an environmental parameter shift θ_t that reshapes the revenue–cost mapping via market competition and league-wide growth effects. We keep the baseline state and control definitions.

7.2 System dynamics with expansion disturbance

We embed expansion into the state transition as:

$$X_{t+1} = AX_t + BU_t - S_{\exp}(t) + \xi_t \tag{23}$$

where ξ_t captures unmodeled noise (zero-mean), and $S_{\exp}(t)$ is a time-varying disturbance vector representing expansion-induced shocks to the three states.

7.3 Time structure of the expansion shock

Expansion typically produces a stronger short-run shock in the first season followed by a weaker but persistent long-run effect.We encode this using a two-regime disturbance:

$$S_{\exp}(t) = \begin{cases} S_{\exp}^{\text{short}}, & t = t_0 (\text{first year of expansion}) \\ S_{\exp}^{\text{long}}, & t \geq t_0 + 1 (\text{steady-state period}) \end{cases} \tag{24}$$

where t_0 denotes the expansion launch year

$$S_{\exp}^{\text{short}}(s) = \begin{cases} S_1^{\text{short}}(s) = -\left(0.03 + 0.05 \cdot \text{TT} + 0.05 \cdot \text{TI}(s)\right) \\ S_2^{\text{short}}(s) = \left(-0.5 \cdot \text{CI}(s) + 0.3 \cdot \text{DP}(s)\right) X_{2,t} \\ S_3^{\text{short}}(s) = 0.385 \end{cases} \tag{25}$$

$$S_{\exp}^{\text{long}}(s) = \begin{cases} S_1^{\text{long}}(s) = 0.05 \cdot \text{TI}(s) X_{1,t} \\ S_2^{\text{long}}(s) = \left(-0.3 \cdot \text{CI}(s) + 0.15 \cdot \text{DP}(s)\right) X_{2,t} \\ S_3^{\text{long}}(s) = -0.133 \end{cases} \tag{26}$$

7.4 Five indices to quantify “location effects” via resource competition

Rather than modeling geography mainly through travel costs, we quantify the impact of expansion locations through resource competition—who competes for the same attention, revenue, and talent. We define five interpretable indices that serve as inputs to $S_{\exp}(t)$ and θ_t.

7.4.1 Competition ImpactMarket (CI)

$$\text{CI}(s) = \begin{cases} 0.75, & s = S1 \\ 0.35, & s = S2 \\ 0.05, & s = S3 \\ 0.40, & s = S4 \end{cases}$$

Domestic-near locations create the strongest substitution; cross-border competition is stronger than a pure distance rule would imply due to institutional, media, and cultural segmentation.

7.4.2 Derby Premium (DP)

$$\mathrm{DP}(s)=\begin{cases}0.50, & s=S1\\ 0.20, & s=S2\\ 0.00, & s=S3\\ 0.05, & s=S4\end{cases}$$

Derby narratives peak in near-distance domestic rivalry, weaken with distance, and remain modest across borders.

7.4.3 League Growth (LG)

$$\mathrm{LG}(s)=\begin{cases}0.10, & s=S1\\ 0.45, & s=S2\\ 0.70, & s=S3\\ 0.90, & s=S4\end{cases}$$

Far-distance and cross-border expansions are assumed to add larger incremental media and sponsorship pools by filling under-served or new markets.

7.4.4 Travel Impact (TI)

Travel effects are encoded with positive values indicating reduced travel burden and negative values indicating increased fatigue and frictions:

$$\mathrm{TI}(s)\in\{+0.15,+0.10,-0.25,-0.35\}\quad \text{for } s\in\{S1,S2,S3,S4\} \tag{27}$$

7.4.5 Talent Tightness (TT)

Expansion increases rotation-slot demand relative to the available rotation-talent pool, yielding a league-wide tightness level:

$$\mathrm{TT}=\frac{2\times 8}{50}=0.32 \tag{28}$$

7.5 Parameter adjustment θ_t: how revenue and costs shift after expansion

We implement θ_t through two channels: (i) revenue reallocation driven by MOI and LGD, and (ii) salary-cost inflation driven by TT. This closes the modeling narrative from "where the new teams are" to "how Fever's business environment changes.

7.5.1 Revenue-side adjustment: CI + DP + LG

The recognition results are shown below:

$$\begin{aligned}\mathrm{Rev}_{t+1}^{\mathrm{local}}&=\mathrm{Rev}_{t}^{\mathrm{local}}\left(1-0.5\cdot\mathrm{CI}(s)+0.3\cdot\mathrm{DP}(s)\right)\\ \mathrm{LeaguePool}_{t+1}&=\mathrm{LeaguePool}_{t}\left(1+0.15\cdot\mathrm{LG}(s)\right)\end{aligned} \tag{29}$$

7.5.2 Cost-side adjustment: TT + TI

We inflate salary-related operating costs via:

$$\begin{aligned}\text{SalaryCost}_{t+1} &= \text{SalaryCost}_t\left(1+0.25\cdot \text{TT}\right)\\ \text{TravelCost}_{t+1} &= \text{TravelCost}_t\left(1-\text{TI}(s)\right)\end{aligned} \tag{30}$$

7.5.3 Optional MPC matrix correction: reduced marginal efficiency of payroll intensity

Because salary inflation lowers the "bang-for-buck" of payroll intensity, we optionally adjust payroll-related effectiveness in the control matrix:

$$B_{1,2}^{new} = \frac{B_{1,2}^{old}}{1+0.25\cdot \text{TT}} \tag{31}$$

7.6 Results and analysis

Across the four expansion-location scenarios, the model produces a stable ranking driven by the CI–DP–LG–TI mechanism: **nearby domestic expansion** has the most negative net impact and pushes down the trajectories of **win rate, profit, and valuation** via stronger local competition, while **far domestic expansion** yields the most positive net impact as league-growth benefits dominate substitution and travel frictions; **mid-distance domestic** and **cross-border** sit in between, and the optimal policy shifts **payroll intensity, star reliance, and marketing** accordingly to buffer downside while capturing upside.

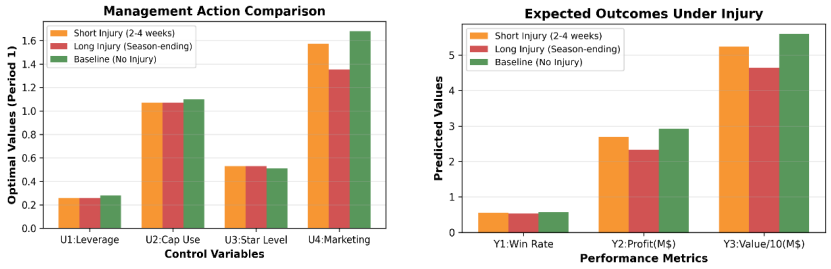


Figure 4: MPC Injury-Response Comparison

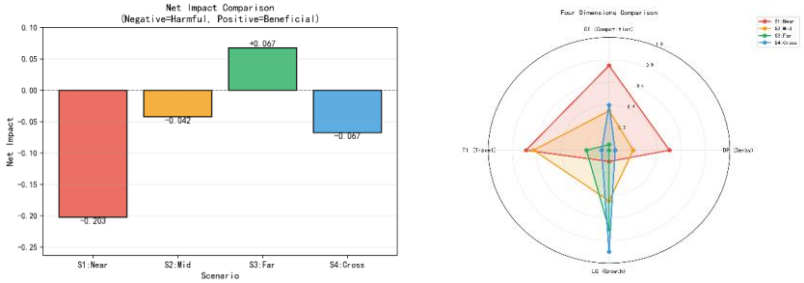


Figure 7: Expansion Scenario Outcomes and Mechanism

8 Task III: Player Injury Shocks

A star-player injury is modeled as a one-time exogenous shock that triggers an immediate state update at the next decision window, after which the MPC re-optimizes a sequence of controllable actions under budget, salary-cap, and roster constraints.

8.1 Shock-based state update

This equation treats a star injury as a one-time external shock that resets the next-window state, where d_t is a 3-dimensional disturbance acting on the three states (competitive strength, fan heat, and financial condition), while ξ_t represents residual noise not captured by the baseline matrices A, B.

$$X_{t+1} = AX_t + BU_t + d_t + \xi_t, \qquad d_t \in \mathbb{R}^3 \tag{32}$$

8.2 Injury impact increments

These increments convert "star-out" into two computable losses—ΔWin for competitive performance and ΔRev for business outcomes—where ΔRev is decomposed into gate/media/merch channels and ΔGateRev is driven mainly by the injury-induced attendance change, scaled by average ticket price and the number of affected home games $N_{\text{home,out}}$

Competitive decrement:

$$\Delta Win = \text{WinRate(roster)} - \text{WinRate(roster without star)} \tag{33}$$

Business decrement:

$$\begin{aligned} &\Delta Rev = \Delta GateRev + \Delta MediaValue + \Delta MerchProxy \\ &\Delta GateRev \approx (\Delta Attendance) \cdot \overline{p} \cdot N_{\text{home,out}} \end{aligned} \tag{34}$$

8.3 MPC re-optimization

After the shock update, the MPC re-optimizes the control trajectory over horizon H by balancing output tracking (weighted by Q) against overly aggressive policy changes (penalized by R on ΔU), so the resulting actions remain feasible under budget/salary-cap/roster constraints while stabilizing Y.

$$\min_{\Delta U_{t:t+H-1}} \sum_{k=1}^{H} \left(Y_{t+k} - W_{t+k}\right)^{\top} Q \left(Y_{t+k} - W_{t+k}\right) + \sum_{k=0}^{H-1} \Delta U_{t+k}^{\top} R \Delta U_{t+k} \tag{35}$$

8.4 Uncertain absence length

We model return-time uncertainty with two absence scenarios $s \in [1,2]$ and probabilities p_s, and choose a plan that maximizes expected performance $\mathbb{E}[\Pi]$ while penalizing risk via $\lambda \text{Var}(\Pi)$, where $\Pi_{t+1}^{(s)}$ is the scenario-specific overall value score and λ controls how conservative the strategy is.

Scenarios:

$$s \in 1,2 \quad \text{(short vs. long absence)}, \qquad \sum_s p_s = 1$$

$$\max \mathbb{E}[\Pi] - \lambda, \mathrm{Var}(\Pi) = \max \sum_s p_s, \Pi_{t+1}^{(s)} - \lambda! \left(\sum_s p_s \left(\Pi_{t+1}^{(s)} \right)^2 - \left(\sum_s p_s, \Pi_{t+1}^{(s)} \right)^2 \right) \quad (36)$$

8.5 Results and analysis

In injury-response simulations, the MPC absorbs the shock by reducing financial exposure and smoothing adjustments—lowering leverage and cap intensity while reallocating marketing (most strongly under long injuries)—but all injury cases still shift the trajectories downward, with the long-injury scenario consistently worst and requiring the highest control/penalty cost under the mean–variance (risk-aware) design.

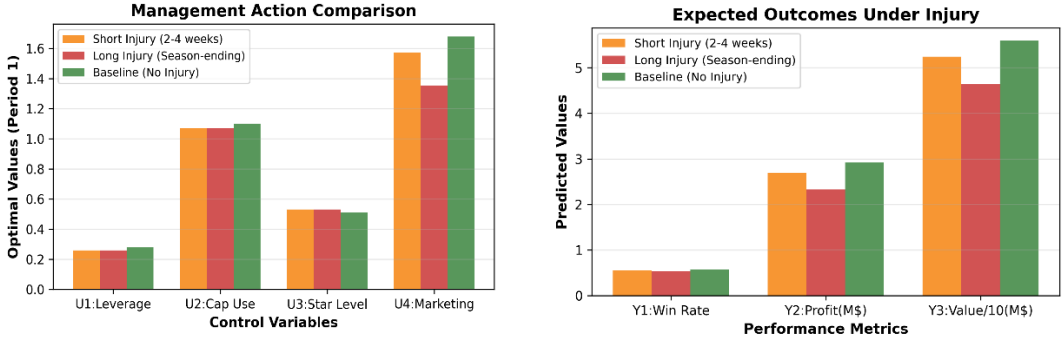


Figure 8: MPC Injury-Response Comparison

9 Task IV: Additional Business Decisions-Ticket Pricing

We incorporate season ticket pricing into the existing MPC by introducing a new control $U_{5,t}$ (normalized average home ticket price factor), and letting it influence (i) attendance demand through a constant price elasticity and (ii) profit/valuation outputs through an explicit ticket-revenue term.

9.1 New control variable and constraints

We define $U_{5,t}$, and adopt the price-elastic demand exponent as follows:

$$U_{5,t} \in [0.7, , 1.3], U_{5,t} = 1.0 \text{ (baseline)}, U_{5,t} > 1 \text{ (price up)}, U_{5,t} < 1 \text{ (price down)}$$
$$\epsilon = -0.8 \quad (37)$$

9.2 Ticket demand and revenue

Ticketing demand and ticketing revenue formulas are defined as follows:

$$\mathrm{Attendance}(t) = A_{\text{base}} \cdot X_{2,t} \cdot U_{5,t}^{\epsilon}$$
$$R_{\text{ticket}}(t) = k_{\text{ticket}} \cdot X_{2,t} \cdot U_{5,t}^{1+\epsilon} = k_{\text{ticket}} \cdot X_{2,t} \cdot U_{5,t}^{0.2}, k_{\text{ticket}} = 2.0 \quad (38)$$

9.3 Augmented state transition

We define an augmented control vector and keep the **no-constant** linear form.

$$\bar{U}_t = \left[U_{1,t}, U_{2,t}, U_{3,t}, U_{4,t}, U_{5,t}\right]^\top$$
$$X_{t+1} = AX_t + \bar{B}, \bar{U}_t$$

The expanded control matrix $\bar{B}$ appends the ticket-price channel:

$$\bar{B} = \begin{bmatrix} 0.03 & 0.42 & 0.08 & 0.01 & 0.00 \\ 0.01 & 0.12 & 0.95 & 0.28 & -0.15 \\ -20.0 & -0.18 & -0.08 & -0.05 & 0.00 \end{bmatrix}$$

9.4 Output reconstruction with $U_{5,t}$

We have modified the calculation formulas for A and B as follows:

$$\begin{aligned} Y_{2,t+1} &= 2X_{2,t}, U_{5,t}^{0.2} + 1.5X_{2,t} - 0.66U_{2,t} + U_{6,t} - 0.05X_{3,t} \\ Y_{3,t+1} &= 20X_{1,t} - 8X_{3,t} + 12U_{2,t} + 24U_{6,t} + 5U_{5,t} \end{aligned} \tag{39}$$

9.5 MPC optimization

After introducing the new variable, the objective function formula is as follows:

$$\min_{\Delta \bar{U}_{t:t+H-1}} \sum_{k=1}^{H} (Y_{t+k} - W_{t+k})^\top Q (Y_{t+k} - W_{t+k}) + \sum_{k=0}^{H-1} \Delta \bar{U}_{t+k}^\top R \Delta \bar{U}_{t+k} \tag{40}$$

9.6 Results and analysis

After solving the optimization problem, we obtain the optimal ticket-pricing strategy as follows:

Keep $U_5 = 1$ in 2026, then raise the average ticket price by +5% each season (2027–2029), i.e., $U_{5,2027} = 1.05, U_{5,2028} = 1.1, U_{5,2029} = 1.15$ .If occupancy rates show a significant decline, pause the next step and recalculate the strategy for the following season based on the current team's status.

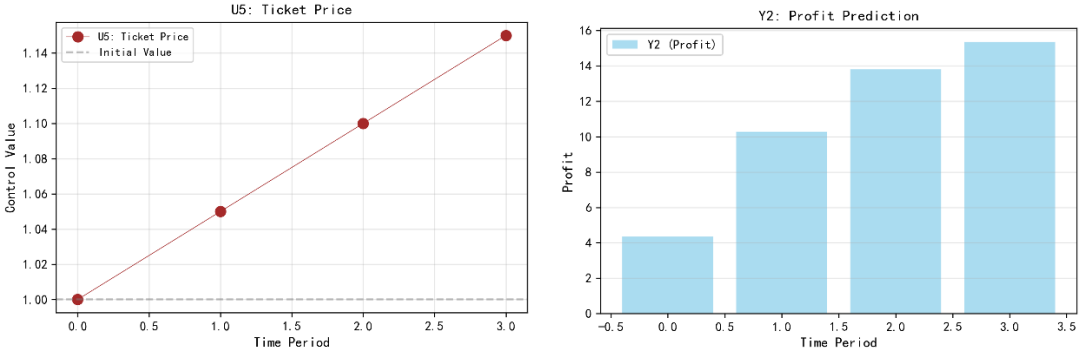


Figure 9: Optimal Ticket-Price Path and Profit Response under MPC

References

[1] Humphreys, B. R., & Mondello, M. (2008). Determinants of franchise values in North American professional sports leagues. International Journal of Sport Finance, 3(1), 40-57.

STRATEGIC ADVISORY LETTER

TO: Herb Simon, Owner of the Indiana Fever
FROM: Team #2605710, Strategic Analytics Division
DATE: February 2, 2026
SUBJECT: Maximizing Franchise Value: A Data-Driven Roadmap for the New WNBA Era



Dear Mr. Simon,

The Indiana Fever is currently at the epicenter of a transformative era for the WNBA. As the league's global visibility reaches unprecedented heights, the traditional management of a sports franchise must transition toward a more integrated commercial strategy. To capitalize on this momentum, our analysis has identified several high-impact strategic pillars designed to ensure the franchise's long-term competitive dominance and financial health.

A primary opportunity for immediate revenue growth lies in the optimization of our ticketing structure. Our analysis of current market demand indicates that fan loyalty for the Fever has become significantly price-inelastic, with a calculated elasticity of approximately -0.8. This suggests that the market is remarkably resilient to price changes in the premium segment. We recommend a bold upward adjustment in premium ticketing and season-pass pricing. Such a move will generate substantial liquidity without risking meaningful attendance drops, providing the necessary capital to compete for elite talent in an increasingly tightening player market.

Beyond direct revenue, the long-term success of the franchise depends on maintaining a precise balance between player acquisition and supporting infrastructure. A common pitfall in professional sports is over-investing in "star power" while neglecting the ecosystem that sustains it. Our findings suggest that for every dollar spent on high-profile player acquisitions, the franchise should allocate approximately 35 cents toward elite training facilities and medical support. Data indicates that failing to maintain this 1:0.35 investment ratio leads to a notable 15% decline in player performance and longevity over just two seasons. In this context, superior infrastructure is a fundamental financial necessity to protect the team's most expensive assets.

Furthermore, the timing of our commercial outreach is as critical as the volume of investment. Marketing and digital engagement exhibit a significant "delayed-action" effect on the franchise's total valuation. We suggest shifting a larger portion of marketing expenditures to the early stages of the season rather than spreading them evenly. Building high digital engagement volume early on acts as a valuation lever, securing premium sponsorship contracts and driving merchandise sell-outs during the postseason when global viewership peaks and the brand's reach is at its maximum.

Finally, given the physical intensity of the WNBA schedule and the inherent volatility of a star-driven league, the franchise must prioritize financial flexibility and risk mitigation. We recommend the establishment of a dedicated Performance Reserve Fund to buffer against unforeseen player injuries. This ensures that commercial operations and marketing momentum remain robust regardless of short-term competitive fluctuations. By combining this with more flexible, incentive-based contract structures, the Indiana Fever will not only dominate on the court but also define the gold standard for professional sports management in this new era.

Sincerely,
Team #2605710 Strategic Analytics Division

Report on Use of AI Tools

1. DeepL (Jan, 2026 version)

Uploaded entire paper written in Mandarin to be translated into English.

2. Claude AI (Anthropic, February 2026 version, Claude Sonnet 4.5)

Query 1: Position-Specific Performance Metrics Recommendation

Input: What are the most representative statistical metrics to evaluate WNBA players in different positions (Guards, Forwards, Centers)? We need metrics that align with each position's tactical responsibilities.

Output: Position-specific metric recommendations:

- **Guard metrics**: Assist-to-turnover ratio, assist share percentage, true shooting percentage
- **Forward metrics**: Paint scoring share, offensive rebound rate, help defense participation rate
- **Center metrics**: Rim protection efficiency, deterrence index (difference in opponent rim attempt rate), paint defense efficiency
- Explanation of why these metrics capture positional value (e.g., guards as playmakers, centers as rim protectors)

Query 2: Nonlinear Modeling of Star Effects

Input : When computing the team's fan-heat index (X_2), how should we quantify the positive externality generated by star players? Should team popularity be approximated by a simple average across players, or should we explicitly account for the disproportionate contribution of top stars?

Output : Proposed Approach.

We recommend incorporating diminishing marginal returns in star effects. Intuitively, the first star delivers the largest incremental boost to attention and demand, while additional stars contribute progressively less due to overlap in audience reach and limited attention capacity. A practical specification is:

- 1st star: full contribution (strong incremental attraction)
- 2nd star: discounted contribution (e.g., 70% of full effect)
- 3rd star: further discounted contribution (e.g., 50% of full effect)
- and so on.

One convenient formulation is:

$$X_2 = X_2^{\text{base}} + \sum_{i=1}^{K} \left(H_i \cdot \omega_i \cdot \delta^{i-1} \right)$$

where X_2^{base} is baseline team heat (e.g., market/team-brand component), H_i denotes the popularity score of the (i)-th ranked star on the roster, ω_i is a scaling weight (optional, e.g., reflecting minutes played or media exposure), and $\delta \in (0,1)$ is the diminishing factor (e.g., ($\delta = 0.7$).

3. OpenAI ChatGPT (Jan, 2026 version)

Auto-completions for code used in preparing our models.